

RAHUL TEMBHARE

Industrial IoT & Full-Stack Developer

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PROFESSIONAL OBJECTIVE

I'm a third-year B.Tech student in Industrial IoT who genuinely enjoys building things — from full-stack web apps to hardware-connected systems. Over the past couple of years I've worked on real projects involving IoT, REST APIs, React, AI pipelines, and voice-controlled smart systems. I'm looking for roles where I can contribute practically, keep learning fast, and work on problems that actually matter.

EDUCATION

B.Tech – Industrial IoT | Priyadarshini College of Engineering, Nagpur

2023 – 2027

Coursework: Embedded Systems, Industrial Automation, Network Protocols, Data Structures, Database Systems | IEEE Member | Hackathon Participant

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++

Backend: FastAPI, Flask – REST APIs, WebSocket communication, OAuth 2.0, async request handling

Frontend: React, HTML, CSS – real-time state management, component-based UI, API integration

IoT & Embedded: ESP32, Raspberry Pi, Arduino – sensor networks, relay control, MQTT, real-time data pipelines

AI / Data: AI Foundations (certified), Generative AI (certified), OpenCV, MediaPipe, data processing & analytics

DevTools: Git, GitHub, VS Code, Notion

PROJECTS

Smart Home Automation System | ESP32 • MQTT • Flutter • FastAPI • Alexa Skills Kit

2026

- Built a full end-to-end smart home platform: ESP32 nodes publish/subscribe over MQTT (Mosquitto broker) to a FastAPI backend, with real-time control via a custom-built Flutter mobile app.
- Integrated Alexa voice control using the Alexa Skills Kit and AWS IoT Core — users can switch devices with voice commands or through the app interchangeably.
- Added physical relay control with a manual override switch so the system stays safe even if the software fails — designed for real-world deployment, not just demos.

Anime Gesture Recognition & Air Drawing App | Python • OpenCV • MediaPipe • React

2026

- Built a real-time hand gesture recognition system using MediaPipe Hands and OpenCV that identifies specific anime-style signs from live webcam input and displays what gesture the user is making — no physical input required.
- Implemented an air drawing feature that tracks fingertip position frame-by-frame and lets users draw freely on screen using only hand movements — strokes persist, can be colour-switched, and cleared with a gesture.
- Wired the Python backend to a React frontend that renders recognised gesture labels and the drawing canvas live, keeping the full pipeline under 100 ms for a smooth, responsive feel.

AI-Powered Web Apps (Beauty & Fitness Analysis) | Python • Flask • OpenCV

2025

- Two Flask apps that take live webcam input, run it through an AI model, and return on-screen feedback. Architecture: capture → process → JSON response → UI update.

Autonomous Robot – Robocraft Competition | Arduino • C++ • Bluetooth

2025

- Built a robot from scratch for a college competition — obstacle detection, motor control, and Bluetooth remote driving. Delivered within a tight budget and timeline.

LEADERSHIP & ACTIVITIES

- **Team Leader** – Smart India Hackathon & TechFest IIT Bombay: assembled teams, divided work, and presented solutions to judges.
- **Workshop Lead** – Ran a Generative AI workshop for 60+ students; wrote content and delivered live coding demos.
- **Volunteer** – Widesoftech Tech Alfa Hackathon (Jan 2026).
- **Member** – IEEE & Forum Committee (Symbiot).
- **Organiser** – College esports events with 100+ participants.
- **Student Mentor** – Taught technical concepts to school students.

CERTIFICATIONS

Udemy – Full-Stack Web Development (Nov 2025) • Deloitte Australia – Data Analytics Job Simulation (Aug 2025) • Artificial Intelligence Foundations • Generative AI